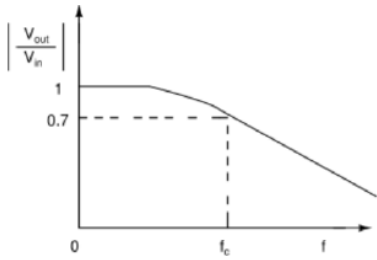


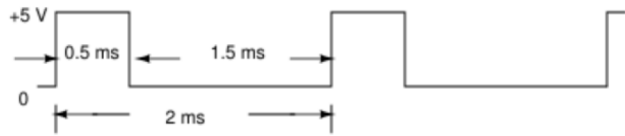
MS101 – Makerspace

2024-25/I Autumn Semester

EE Midsemester Examination: Sep 19, 2024

Answers to Objective and Numerical Questions

Ques.	Answers	Comments
1	B, C, E	
2	A, F	
3	$I_S = 5.25 \text{ mA}$	
4	$V_X = 3 \text{ V}$; $I_L = 5 \text{ mA}$	
5	A) $f_c = 3978.87 \text{ Hz}$. (allowed range: 3977 to 3982) B) A for $f = f_c$: 0.707 (= 0.7) C) A for $f \ll f_c$: 1.0 D)  (Bode plot with log f as the X-axis also acceptable)	
6	A) 6.75 V (allowed range: 6.74 to 6.76) B) 21.21 V C) 10.61 V	
7	A) - 0.25 B) - 40 C) 10 D) Yes	
8	A) 3.5 B) 40 kΩ C) - 5 D) 20 kΩ	
9	Minimum decimal number = -16 Maximum decimal number = +15	
10	1E1	

11	<table><tr><td>AB↓ CD→</td><td>00</td><td>01</td><td>11</td><td>10</td></tr><tr><td>00</td><td>1</td><td></td><td></td><td></td></tr><tr><td>01</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>11</td><td></td><td></td><td></td><td></td></tr><tr><td>10</td><td>1</td><td></td><td></td><td></td></tr></table> $F_n = \bar{A} B + \bar{B} \bar{C} \bar{D}$	AB↓ CD→	00	01	11	10	00	1				01	1	1	1	1	11					10	1				
AB↓ CD→	00	01	11	10																							
00	1																										
01	1	1	1	1																							
11																											
10	1																										
12	Out = $A C + B \bar{C}$																										
13	A) 100 B) 100011																										
14	<table><tr><td>CLK</td><td>Q1</td><td>Q0</td><td>X</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>2</td><td>1</td><td>1</td><td>0</td></tr><tr><td>3</td><td>0</td><td>1</td><td>1</td></tr><tr><td>4</td><td>1</td><td>0</td><td>1</td></tr></table>	CLK	Q1	Q0	X	0	0	1	1	1	1	0	1	2	1	1	0	3	0	1	1	4	1	0	1		
CLK	Q1	Q0	X																								
0	0	1	1																								
1	1	0	1																								
2	1	1	0																								
3	0	1	1																								
4	1	0	1																								
15	$I_{DUT} = -1.75 \text{ mA.}$ $V_{DUT} = -4.5 \text{ V.}$ $V_{out} = +3.5 \text{ V.}$																										
16	A) 199 or 200 B) 99 or 100																										
17	 A) B) 1.25 V																										
18	B, D, E																										